Implantació de tècniques d'accés remot. OpenVPN

Mohamed Serroukh

ÍNDEX

Entitat certificadora: **2**
Certificat servidor: **4**
Firmar certificat: **6**
Diffie-Hellman (DH) **7**
Resum: **7**
TLS: **7**
Configuració: **7**
Certificats client: **9**
Configuració .ovpn: **11**
● Comprovar l'accés i veure que la comunicació està xifrada (Wireshark) **11**
Evidències per lliurar: **12**

- Instal·lar el servidor en una màquina Linux

En aquesta captura podem veure la instal·lació dels paquets **openvpn** i **easy-rsa**

```
root@apache:/home/mserroukh# sudo apt install openvpn easy-rsa -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
```

Entitat certificadora:

Aquí podem veure:

- **Preparació:** Es crea i s'entra al directori de treball (`openvpn-ca`).
- **Inicialització:** S'executa `./easyrsa init-pki`, que **crea l'estructura de carpetes necessària** per gestionar els certificats i les claus de seguretat.

```
root@apache:/# make-cadir ~/openvpn-ca
root@apache:/# cd ~/openvpn-ca
root@apache:~/openvpn-ca# ls
easyrsa  openssl-easyrsa.cnf  vars  x509-types
root@apache:~/openvpn-ca# ./easyrsa init-pki

Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /root/openvpn-ca/pki

Using Easy-RSA configuration:
* /root/openvpn-ca/vars

root@apache:~/openvpn-ca#
```

Aquí s'ha generat el certificat arrel (ca.crt), que és el fitxer mestre que s'utilitzarà per signar i validar tots els certificats de servidor i de clients en la xarxa VPN.

- Comanda executada: `./easyrsa build-ca nopass`. L'atribut `nopass` indica que la clau privada de la CA s'ha creat sense contrasenya.
- Paràmetres de configuració: S'ha definit el Common Name (CN) com a `mohamedvpn`, que actua com l'identificador únic de l'autoritat certificadora.
- Resultat: El sistema confirma que el procés ha finalitzat correctament i ha desat el certificat a la ruta `/root/openvpn-ca/pki/ca.crt`.

```
root@apache:~/openvpn-ca# ./easyrsa build-ca nopass
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
.....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*.+.....+.......+......+
.....+...+..........+......+........++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+++++++*..+......+.....+..........+...................+...+........+.......+..+...+....+......+
..+..........+...+.....+.......+..+.+...............+.........+....+.....+.+.....+.+...+..+....
.....+.+.....+.+..+...+..........+..+..........+.....................+...............+......+.+
...+...+........+.+.....+...................+............+..+...+.......+..+....+...........+.
.....+............+..+...................+........+.+...........+......+....+.....+...+......
....+...+.....+.........+.+..+..........+......+..+...+.+++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++
.+.....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*..+...+++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++*.+......+...+........+...+....+......+..+..
..+...+...+..+.......+.........+..+....+......+...........++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:mohamedvpn

Notice
------
CA creation complete. Your new CA certificate is at:
* /root/openvpn-ca/pki/ca.crt

root@apache:~/openvpn-ca#
```

Certificat servidor:

A aquesta imatge es documenta la preparació de les credencials específiques per al servidor OpenVPN.

- **Comanda:** `./easyrsa gen-req server nopass`.
- **Acció:** Es generen dos fitxers fonamentals:
    1. Clau Privada (`server.key`): La clau secreta que farà servir el servidor per xifrar el trànsit.
    2. Sol·licitud de Signatura de Certificat (`server.req`): Un fitxer que conté la clau pública i la identitat del servidor, pendent de ser signat per la CA creada anteriorment.

  Resultat: Els fitxers s'emmagatzemen a les rutes de la PKI: `/pki/private/server.key` i `/pk`

```
root@apache:~/openvpn-ca# ./easyrsa gen-req server nopass
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
.+.............+..+.+.....+.............+......+...............+...........+....+.........+...
...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*.....+.++++++++++++++++++
+++++++++++++++++++++++++++++++++++++++++++++++++*...+..+.........+...+......+.......+...+..+...
.+...+........+...+.......+...............+.+...+.............+.......................+.....
+.+...............+..+............+.......+......+........+....+...+..+.+......+..............
+.......+............+.....+.......+.....+.+..+.+............+.....+.......+...+............+.
.....+..+.........+......+..........+.....+..........+..+.......+.....+...+.+.........+.......
........+..+....+.....+....+.......................+....+......+..+...+....+...+...........+.+
.....+...+......+....+...+........+.+...+..................+..+.+..+..................+....+..
.+............+......+...........+...+.......+.....+.+................+...+..+..........+..+.+.
.....+...+............+........+.+.....+......+.........+.......+...+..............+..........
+...+.....+.......+...+..+...+.......+..+...+......+.......+...+.....+.+.....+.+..............
+...+......+....+.....+...+.............+.....+......+.............+......+..+...+.+......+...
+..+....+.....+..........+.....+...+.+............+..+......+....+..+.+..................+.+...
...+...+.....+.+.....+.+...........+....+........+......+.........+......+.+........+...+....+
...+..+....+.....................+...+.....+...+..................+....+......+.....+......+.+
.........+.....+.........+......+....+..+++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++
.+....+.....+......+...+....+...+............+..+......++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++*..+....+.........+......++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++*......+...+..+...+..........+..+.......+........+.+...........+.+.....+
....+...+..+............+..+....+..+.......+..+....+..+.......+...+.....+...+....+......+....
.+......+.+...+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [server]:mohamedvpn

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /root/openvpn-ca/pki/reqs/server.req
* key: /root/openvpn-ca/pki/private/server.key

root@apache:~/openvpn-ca#
```

Firmar certificat:

La captura mostra la **signatura amb èxit d'un certificat de servidor** per a una VPN (OpenVPN) mitjançant la utilitat **Easy-RSA**.

- **Acció:** S'ha firmat la sol·licitud (CSR) per a l'entitat `mohamedvpn`.
- **Resultat:** S'ha generat el fitxer final a `/root/openvpn-ca/pki/issued/server.crt`.
- **Caducitat:** El certificat serà vàlid durant **825 dies** (fins al maig de 2028).
- **Estat:** La base de dades de la CA s'ha actualitzat correctament.

```
root@apache:~/openvpn-ca# ./easyrsa sign-req server server
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a server certificate
for '825' days:

subject=
    commonName                = mohamedvpn

Type the word 'yes' to continue, or any other input to abort.
  Confirm request details: yes

Using configuration from /root/openvpn-ca/pki/openssl-easyrsa.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'mohamedvpn'
Certificate is to be certified until May 21 17:12:13 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
------
Certificate created at:
* /root/openvpn-ca/pki/issued/server.crt
```

Diffie-Hellman (DH)

Aquesta captura mostra la generació dels paràmetres Diffie-Hellman (DH), un pas crucial per a la seguretat de la teva VPN.

Resum:

- **Comanda:** S'ha executat `./easyrsa gen-dh`.
- **Acció:** El sistema està creant una clau de seguretat basada en un "safe prime" de 2048 bits.
- **Objectiu:** Aquests paràmetres s'utilitzen per a l'intercanvi de claus xifrades entre el servidor i el client, garantint que ningú pugui interceptar la comunicació.

```
root@apache:~/openvpn-ca# ./easyrsa gen-dh
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
Generating DH parameters, 2048 bit long safe prime
.....................................................................................
.....................................................+........+.........................
.....................................................................+...................+.
.....................................................+...+.........................
.......................+...............................+....................................
.....................................................................................
.....................................................................................
.....................................................................................
.....................................................................+...............
.....................................................................................
.....................................................................................+........
...............+.....................................................................+..........
.....................................+...............................................
.....................................................................................
.....................................................+.......................+....
.....................................................................................
.....................................................................+...............
.....................................................................................
```

TLS:

Aqui generem una clau compartida (HMAC) que afegeix una capa de seguretat extra anomenada TLS Authentication.

Ajuda a protegir el servidor contra atacs de denegació de servei (DoS) i escanejos de ports, ja que el servidor ignorarà qualsevol paquet que no estigui firmat amb aquesta clau.

```
root@apache:~/openvpn-ca# openvpn --genkey secret ta.key
```

Configuració:

Aqui copiem tots els fitxers (`ca.crt`, `server.crt`, `server.key`, `dh.pem` i `ta.key`) al directori actiu d'OpenVPN: `/etc/openvpn/server/`.

```
root@apache:~/openvpn-ca# sudo cp pki/ca.crt pki/issued/server.crt pki/private/server.key pki/
dh.pem ta.key /etc/openvpn/server/
```

```
  GNU nano 7.2                                  server.conf *
port 1194
proto udp
dev tun
ca server/ca.crt
cert server/server.crt
key server/server.key
dh server/dh.pem
auth SHA256
tls-auth server/ta.key 0
server 10.8.0.0 255.255.255.0
ifconfig-pool-persist ipp.txt
keepalive 10 120
cipher AES-256-CBC
user nobody
group nogroup
persist-key
persist-tun
status openvpn-status.log
verb 3
```

`systemctl start`: Posa en marxa el servidor immediatament.

`systemctl enable`: Configura el servidor perquè s'engegui automàticament cada vegada que es reiniciï la màquina.

```
root@apache:/etc/openvpn# sudo systemctl start openvpn@server
root@apache:/etc/openvpn# sudo systemctl enable openvpn@server
Created symlink /etc/systemd/system/multi-user.target.wants/openvpn@server.service → /usr/lib/
systemd/system/openvpn@.service.
```

- Instal·lar un client en una màquina Windows o Linux.

Certificats client:

Aquí creem el certificat del client.

```
root@apache:~/openvpn-ca# ./easyrsa gen-req client1 nopass
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
.+.......+......+.....+...+.......+...........++++++++++++++++++++++++++++++++++++++++++++++++
+++++++++++++++++*.........+.+.....+.+.............+.......++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++*...+...........+..........+..+...+.+.........+...+........+.
+......+........+.+.....+...+.+..+...+.....................................+..........+.....+..
..+..+...+.+...........+.......+..+.+...+......+.....+....+..+..........+..+......+.+...+...+.
....+.......+........+....+........+......+.+......+.....+....+........+.+.....+..........+...
..+......+...+...+...+.......+...........+..+....+...........+................+........+....+
..+..................+......+......+...+......+....+......+..+...+......+....+...........+....
.........++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
.+...+...........+......+...+.......+.....+....+..+...........+...+...+.+........+......+....
.....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*.+.+..............+....
+......+........+.......+...+........+...+...............++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++*....+...+.+...+...+.....+.......+...+.....+....+...+.....+...+...
.+......++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [client1]:client1

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /root/openvpn-ca/pki/reqs/client1.req
* key: /root/openvpn-ca/pki/private/client1.key
```

Aquí signem el certificat del client.

```
root@apache:~/openvpn-ca# ./easyrsa sign-req client client1
Using Easy-RSA 'vars' configuration:
* /root/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a client certificate
for '825' days:

subject=
    commonName                = client1

Type the word 'yes' to continue, or any other input to abort.
  Confirm request details: yes

Using configuration from /root/openvpn-ca/pki/openssl-easyrsa.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'client1'
Certificate is to be certified until May 21 17:45:58 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
------
Certificate created at:
* /root/openvpn-ca/pki/issued/client1.crt
```

Aquest son els arxius que hem de tenir al client:

```
root@ubuntu-virtualbox:/home/ubuntu/vpn# ls
ca.crt  client1.crt  client1.key  client1.ovpn  ta.key
```

Configuració .ovpn:

```
  GNU nano 7.2                                  client1.ovpn *
client
dev tun
proto udp
remote 10.0.2.4 1194
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
auth SHA256
cipher AES-256-CBC

ca ca.crt
cert client1.crt
key client1.key
tls-auth ta.key 1

key-direction 1
verb 3
```

- Comprovar l'accés i veure que la comunicació està xifrada (**Wireshark**)

Aquí ens connectem des del client:

```
root@ubuntu-virtualbox:/home/ubuntu/vpn# sudo openvpn --config client1.ovpn
2026-03-01 14:00:46 DEPRECATED OPTION: --cipher set to 'AES-256-CBC' but missing in --data-ciphers (AES-256-GCM:AE
S-128-GCM:CHACHA20-POLY1305). OpenVPN ignores --cipher for cipher negotiations.
2026-03-01 14:00:46 Note: Kernel support for ovpn-dco missing, disabling data channel offload.
2026-03-01 14:00:46 OpenVPN 2.6.14 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [
AEAD] [DCO]
2026-03-01 14:00:46 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-03-01 14:00:46 DCO version: N/A
2026-03-01 14:00:46 TCP/UDP: Preserving recently used remote address: [AF_INET]10.0.2.4:1194
2026-03-01 14:00:46 Socket Buffers: R=[212992->212992] S=[212992->212992]
2026-03-01 14:00:46 UDPv4 link local: (not bound)
2026-03-01 14:00:46 UDPv4 link remote: [AF_INET]10.0.2.4:1194
2026-03-01 14:00:46 TLS: Initial packet from [AF_INET]10.0.2.4:1194, sid=a87a750b 4d9f74f7
2026-03-01 14:00:46 VERIFY OK: depth=1, CN=mohamedvpn
2026-03-01 14:00:46 VERIFY KU OK
2026-03-01 14:00:46 Validating certificate extended key usage
2026-03-01 14:00:46 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authenticat
ion
2026-03-01 14:00:46 VERIFY EKU OK
2026-03-01 14:00:46 VERIFY OK: depth=0, CN=mohamedvpn
2026-03-01 14:00:46 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits R
SA, signature: RSA-SHA256, peer temporary key: 253 bits X25519
2026-03-01 14:00:46 [mohamedvpn] Peer Connection Initiated with [AF_INET]10.0.2.4:1194
2026-03-01 14:00:46 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
```

SI fem un ip a podrem veure el tunnel:

```
4: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.6 peer 10.8.0.5/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::b144:6639:ddd1:defc/64 scope link stable-privacy
       valid_lft forever preferred_lft forever
```

Evidències per lliurar:

- Fitxers de configuració del servidor (`/etc/openvpn/server.conf`) i del client (`client1.ovpn`).

```
  GNU nano 7.2                                          client1.ovpn
client
dev tun
proto udp
remote 10.0.2.4 1194
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
auth SHA256
cipher AES-256-CBC

ca ca.crt
cert client1.crt
key client1.key
tls-auth ta.key 1

key-direction 1
verb 3
```

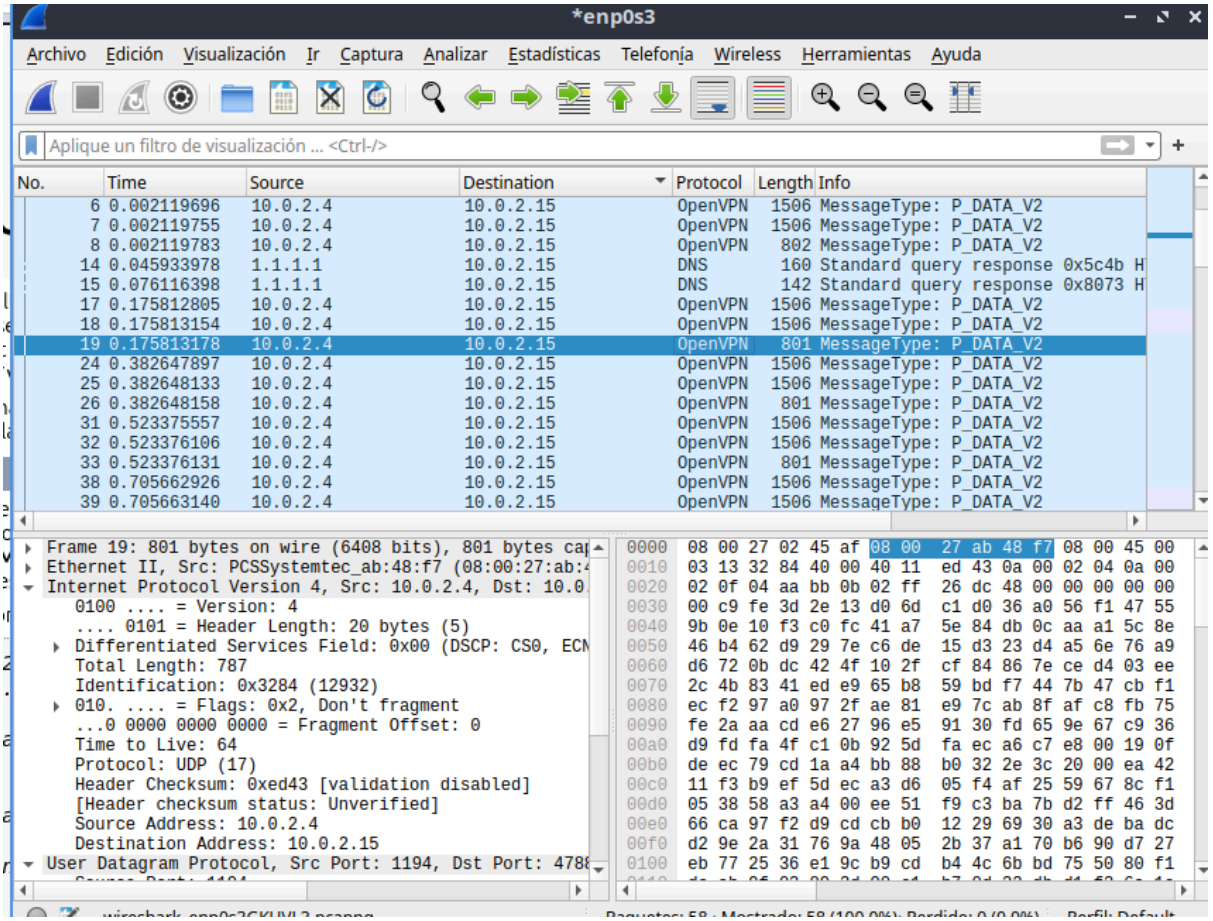
```
  GNU nano 7.2                                          server.conf
port 1194
proto udp
dev tun
ca server/ca.crt
cert server/server.crt
key server/server.key
dh server/dh.pem
auth SHA256
tls-auth server/ta.key 0
server 10.8.0.0 255.255.255.0
ifconfig-pool-persist ipp.txt
keepalive 10 120
cipher AES-256-CBC
user nobody
group nogroup
persist-key
persist-tun
status openvpn-status.log
verb 3
```

- Logs d'arrencada del servei OpenVPN (`journalctl -u openvpn@server`).

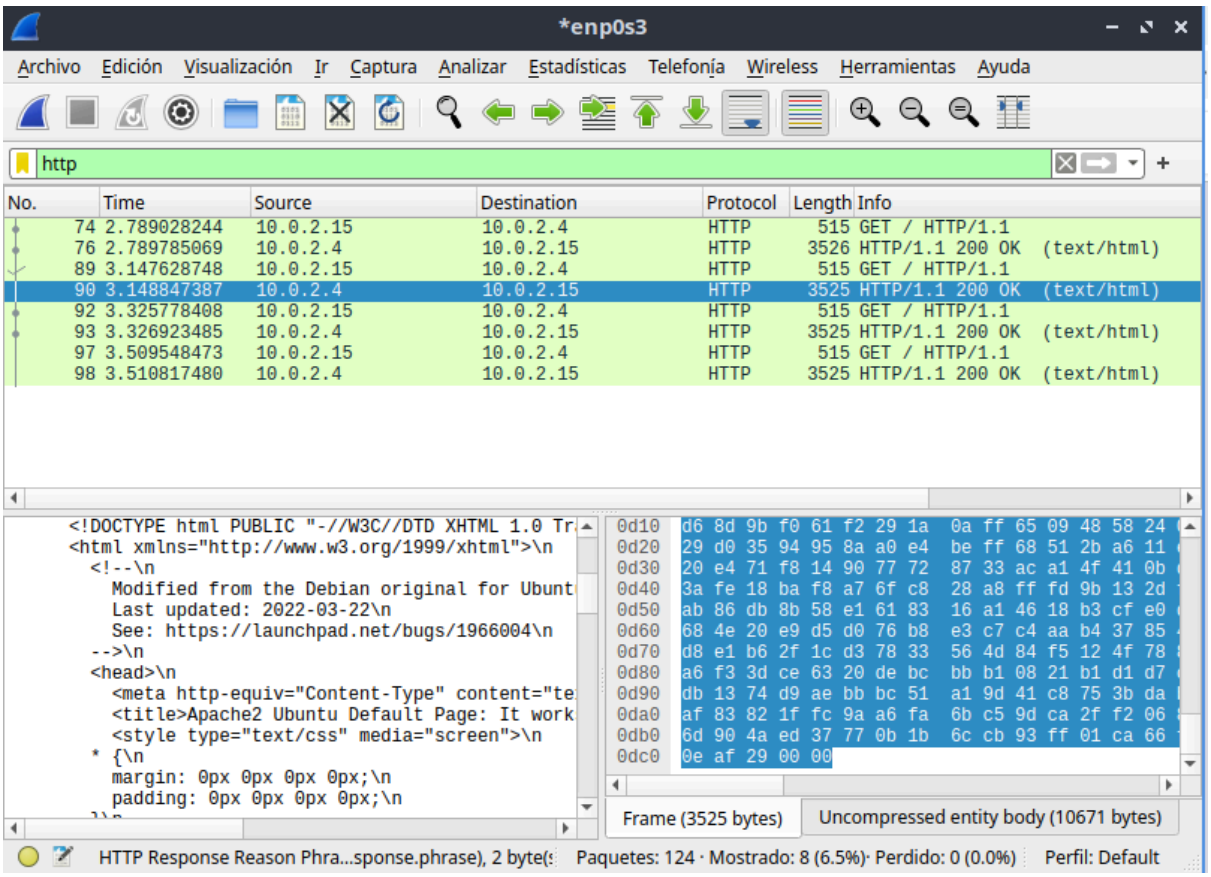
```
feb 16 17:37:22 apache systemd[1]: Starting openvpn@server.service - OpenVPN connection to server...
feb 16 17:37:22 apache ovpn-server[7140]: WARNING: --topology net30 support for server configs with IPv4 pools will be removed in a future release. Please migr
feb 16 17:37:22 apache ovpn-server[7140]: DEPRECATED OPTION: --cipher set to 'AES-256-CBC' but missing in --data-ciphers (AES-256-GCM:AES-128-GCM:CHACHA20-POLY
feb 16 17:37:22 apache ovpn-server[7140]: --user specified but lacking CAP_SETPCAP. Cannot retain CAP_NET_ADMIN. Disabling data channel offload
feb 16 17:37:22 apache ovpn-server[7140]: OpenVPN 2.6.14 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
feb 16 17:37:22 apache ovpn-server[7140]: library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
feb 16 17:37:22 apache ovpn-server[7140]: DCO version: N/A
feb 16 17:37:22 apache systemd[1]: Started openvpn@server.service - OpenVPN connection to server.
feb 16 17:37:22 apache ovpn-server[7140]: net_route_v4_best_gw query: dst 0.0.0.0
feb 16 17:37:22 apache ovpn-server[7140]: net_route_v4_best_gw result: via 10.0.2.1 dev enp0s3
feb 16 17:37:22 apache ovpn-server[7140]: Diffie-Hellman initialized with 2048 bit key
feb 16 17:37:22 apache ovpn-server[7140]: net_route_v4_best_gw query: dst 0.0.0.0
feb 16 17:37:22 apache ovpn-server[7140]: net_route_v4_best_gw result: via 10.0.2.1 dev enp0s3
feb 16 17:37:22 apache ovpn-server[7140]: ROUTE_GATEWAY 10.0.2.1/255.255.255.0 IFACE=enp0s3 HWADDR=08:00:27:ab:48:f7
feb 16 17:37:22 apache ovpn-server[7140]: TUN/TAP device tun0 opened
feb 16 17:37:22 apache ovpn-server[7140]: net_iface_mtu_set: mtu 1500 for tun0
feb 16 17:37:22 apache ovpn-server[7140]: net_iface_up: set tun0 up
feb 16 17:37:22 apache ovpn-server[7140]: net_addr_ptp_v4_add: 10.8.0.1 peer 10.8.0.2 dev tun0
feb 16 17:37:22 apache ovpn-server[7140]: net_route_v4_add: 10.8.0.0/24 via 10.8.0.2 dev [NULL] table 0 metric -1
feb 16 17:37:22 apache ovpn-server[7140]: Could not determine IPv4/IPv6 protocol. Using AF_INET
feb 16 17:37:22 apache ovpn-server[7140]: Socket Buffers: R=[212992->212992] S=[212992->212992]
feb 16 17:37:22 apache ovpn-server[7140]: UDPv4 link local (bound): [AF_INET][undef]:1194
feb 16 17:37:22 apache ovpn-server[7140]: UDPv4 link remote: [AF_UNSPEC]
feb 16 17:37:22 apache ovpn-server[7140]: UID set to nobody
feb 16 17:37:22 apache ovpn-server[7140]: GID set to nogroup
feb 16 17:37:22 apache ovpn-server[7140]: Capabilities retained: CAP_NET_ADMIN
feb 16 17:37:22 apache ovpn-server[7140]: MULTI: multi_init called, r=256 v=256
feb 16 17:37:22 apache ovpn-server[7140]: IFCONFIG POOL IPv4: base=10.8.0.4 size=62
feb 16 17:37:22 apache ovpn-server[7140]: IFCONFIG POOL LIST
feb 16 17:37:22 apache ovpn-server[7140]: Initialization Sequence Completed
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 VERIFY OK: depth=1, CN=mohamedvpn
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 VERIFY OK: depth=0, CN=client1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_VER=2.6.14
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_PLAT=linux
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_TCPNL=1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_MTU=1600
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_NCP=2
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_CIPHERS=AES-256-GCM:AES-128-GCM:CHACHA20-POLY1305
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_PROTO=990
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_LZO_STUB=1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_COMP_STUB=1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 peer info: IV_COMP_STUBv2=1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 TLS: tls_multi_process: initial untrusted session promoted to trusted
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, sig
feb 16 18:06:32 apache ovpn-server[7140]: 10.0.2.15:52979 [client1] Peer Connection Initiated with [AF_INET]10.0.2.15:52979
feb 16 18:06:32 apache ovpn-server[7140]: client1/10.0.2.15:52979 MULTI_sva: pool returned IPv4=10.8.0.6, IPv6=(Not enabled)
feb 16 18:06:32 apache ovpn-server[7140]: client1/10.0.2.15:52979 MULTI: Learn: 10.8.0.6 -> client1/10.0.2.15:52979
feb 16 18:06:32 apache ovpn-server[7140]: client1/10.0.2.15:52979 MULTI: primary virtual IP for client1/10.0.2.15:52979: 10.8.0.6
lines 1-49
```

- Captures Wireshark/tcpdump:

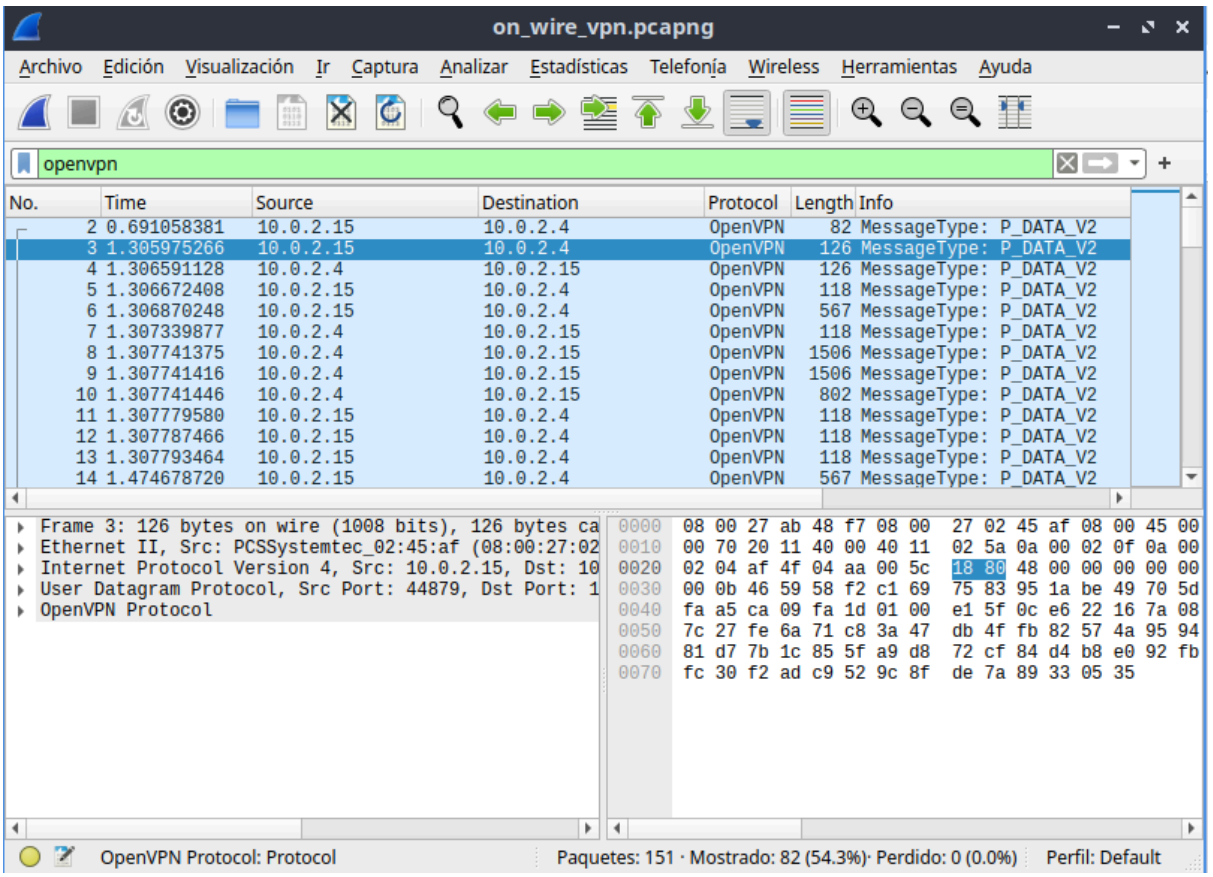
`before_vpn.pcap`



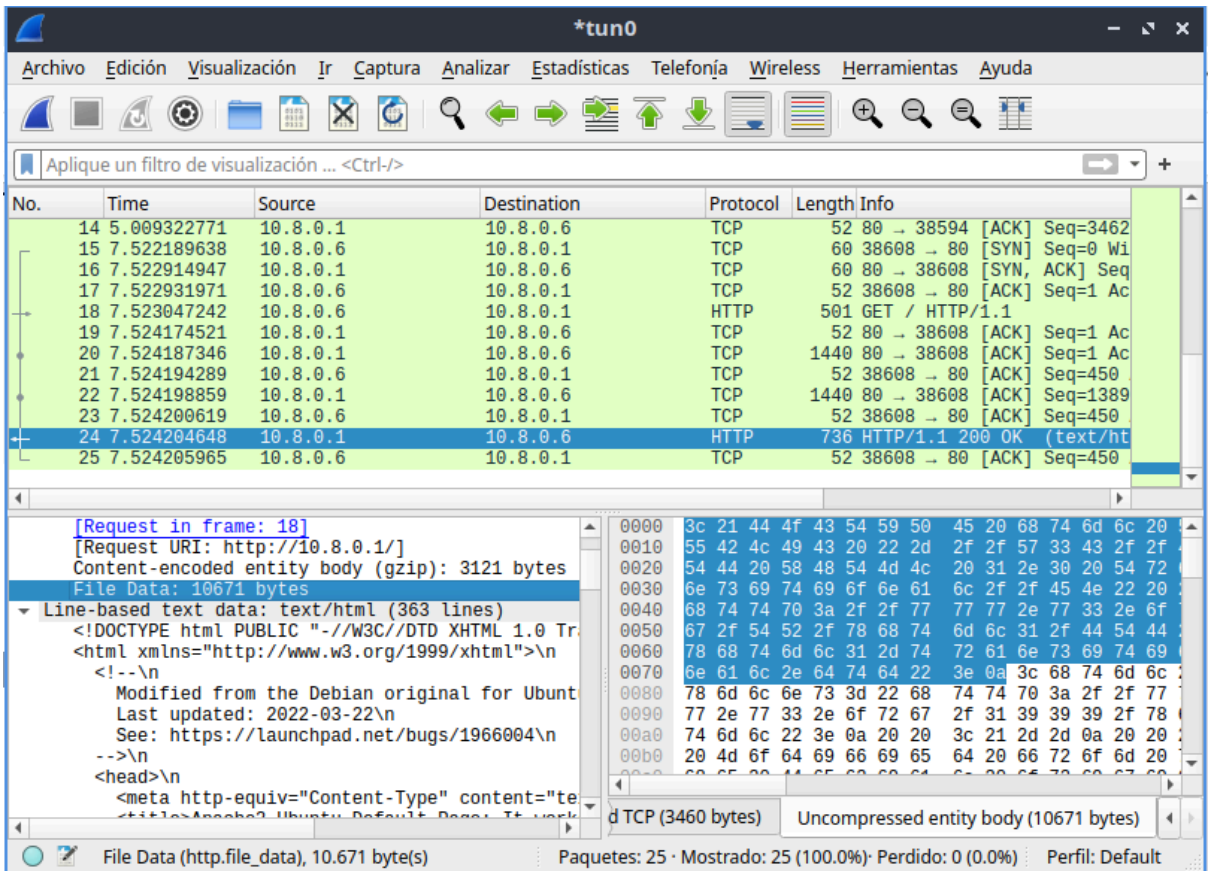
- (client accedint a Apache directament; HTTP en clar).



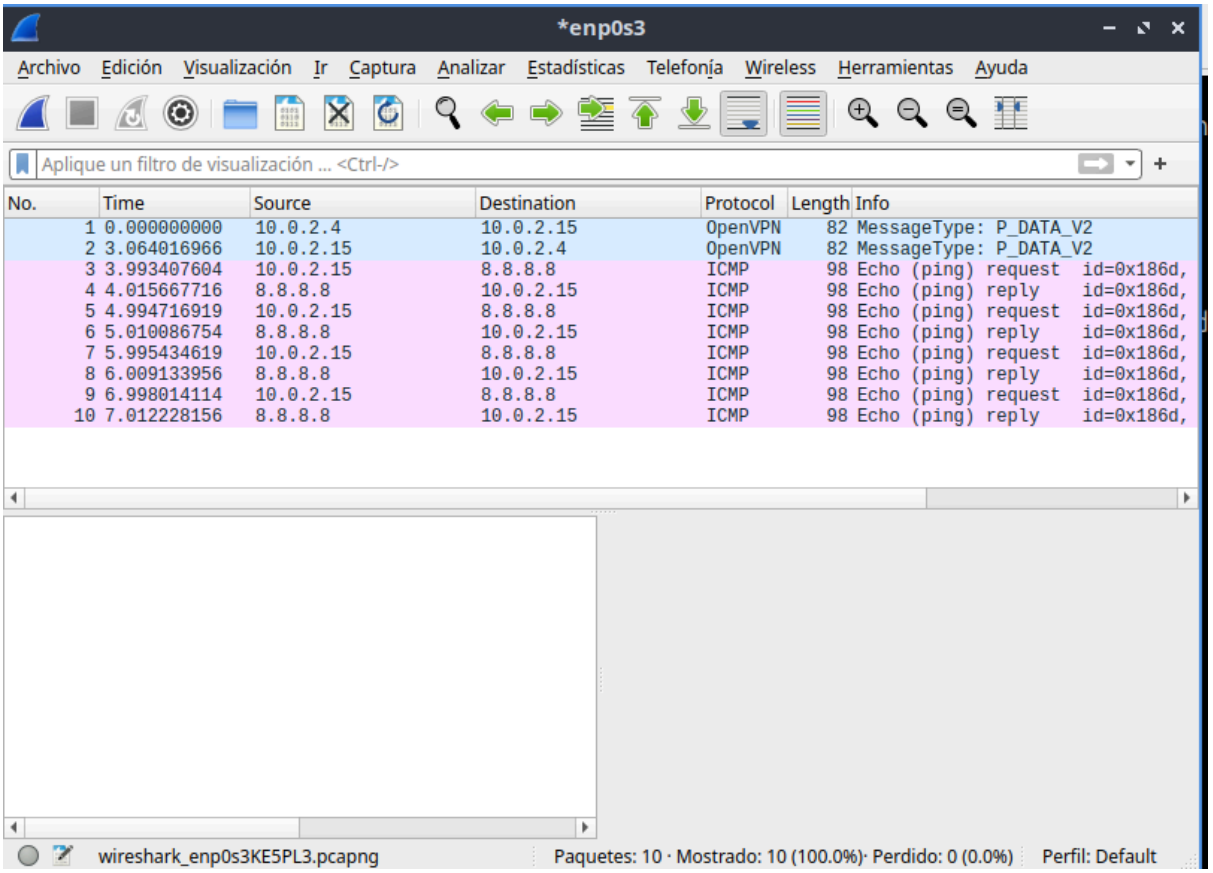
- `on_wire_vpn.pcap` (captura sobre interfície física amb VPN connectada; només paquets OpenVPN).



- `tun_vpn.pcap` (captura a `tun0` mostrant trànsit interior



- comprovacions curl/ping amb sortida i qualsevol captura de pantalla o sortida de comandes per verificar connectivitat i xifrat.


*enp0s3
Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda
Aplique un filtro de visualización ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.2.4 10.0.2.15 OpenVPN 82 MessageType: P_DATA_V2
2 3.064016966 10.0.2.15 10.0.2.4 OpenVPN 82 MessageType: P_DATA_V2
3 3.993407604 10.0.2.15 8.8.8.8 ICMP 98 Echo (ping) request id=0x186d,
4 4.015667716 8.8.8.8 10.0.2.15 ICMP 98 Echo (ping) reply id=0x186d,
5 4.994716919 10.0.2.15 8.8.8.8 ICMP 98 Echo (ping) request id=0x186d,
6 5.010086754 8.8.8.8 10.0.2.15 ICMP 98 Echo (ping) reply id=0x186d,
7 5.995434619 10.0.2.15 8.8.8.8 ICMP 98 Echo (ping) request id=0x186d,
8 6.009133956 8.8.8.8 10.0.2.15 ICMP 98 Echo (ping) reply id=0x186d,
9 6.998014114 10.0.2.15 8.8.8.8 ICMP 98 Echo (ping) request id=0x186d,
10 7.012228156 8.8.8.8 10.0.2.15 ICMP 98 Echo (ping) reply id=0x186d,
wireshark_enp0s3KE5PL3.pcapng
Paquetes: 10 · Mostrado: 10 (100.0%)· Perdido: 0 (0.0%)
Perfil: Default